CLASS 10 — SCIENCE · FORMULA SHEET

Chapter 3: **Metals and Non-metals** | Quick revision, one page

Properties

- Metals: lustrous, malleable, ductile, good conductors of heat and electricity, sonorous, usually hard with high melting points (exceptions: Na and K soft, Hg liquid, Ga and Cs melt in the hand).
- Non-metals: dull, brittle, poor conductors (graphite conducts), low melting points (diamond is the hardest natural substance; iodine is lustrous).

Reactions of metals

- Metal + oxygen $\rightarrow$ metal oxide: $4\,\mathrm{Al} + 3\,\mathrm{O_2} \longrightarrow 2\,\mathrm{Al_2O_3}$. Na and K are stored under kerosene.
- Amphoteric oxides ($\mathrm{Al_2O_3}$, ZnO): $\mathrm{Al_2O_3} + 6\,\mathrm{HCl} \longrightarrow 2\,\mathrm{AlCl_3} + 3\,\mathrm{H_2O}$ and $\mathrm{Al_2O_3} + 2\,\mathrm{NaOH} \longrightarrow 2\,\mathrm{NaAlO_2} + \mathrm{H_2O}$.
- Water: $2\,\mathrm{Na} + 2\,\mathrm{H_2O} \longrightarrow 2\,\mathrm{NaOH} + \mathrm{H_2}$; steam: $3\,\mathrm{Fe} + 4\,\mathrm{H_2O} \longrightarrow \mathrm{Fe_3O_4} + 4\,\mathrm{H_2}$ (Mg, Al, Zn, Fe react with steam).
- Dilute acid: metal + acid $\rightarrow$ salt + $\mathrm{H_2}$ (Cu, Ag, Au, Hg do not displace hydrogen).
- Displacement: $\mathrm{Fe} + \mathrm{CuSO_4} \longrightarrow \mathrm{FeSO_4} + \mathrm{Cu}$ (more reactive metal displaces less reactive).

Reactivity series (decreasing)

- K > Na > Ca > Mg > Al > Zn > Fe > Pb > [H] > Cu > Hg > Ag > Au.

Ionic compounds

- Formed by transfer of electrons: $\mathrm{Na} \longrightarrow \mathrm{Na^+} + \mathrm{e^-}$; $\mathrm{Cl} + \mathrm{e^-} \longrightarrow \mathrm{Cl^-}$; Mg with 2, 8, 2 gives $\mathrm{Mg^{2+}}$.
- Properties: solid, hard and brittle, high melting and boiling points, soluble in water, conduct only in molten state or in solution.

Metallurgy

- Steps: concentration of ore $\rightarrow$ extraction of metal $\rightarrow$ refining.
- Low reactivity (Hg): $2\,\mathrm{HgS} + 3\,\mathrm{O_2} \longrightarrow 2\,\mathrm{HgO} + 2\,\mathrm{SO_2}$, then $2\,\mathrm{HgO} \longrightarrow 2\,\mathrm{Hg} + \mathrm{O_2}$. Middle (Zn, Fe, Pb): roast/calcine, then reduce with carbon (e.g. $\mathrm{ZnO} + \mathrm{C} \longrightarrow \mathrm{Zn} + \mathrm{CO}$).
- High reactivity (Na, Ca, Al): electrolysis of molten chloride/oxide; cathode gets the metal. Thermit: $\mathrm{Fe_2O_3} + 2\,\mathrm{Al} \longrightarrow 2\,\mathrm{Fe} + \mathrm{Al_2O_3}$ + heat.
- Refining: electrolytic refining (anode impure metal, cathode pure metal, electrolyte a salt solution of the metal).

Corrosion

- Rusting needs air and moisture; prevented by painting, oiling, galvanising, alloying. Silver turns black, copper gets a green coating. Alloy = homogeneous mixture of metals (brass, bronze, solder).